

CSE 422 Lab Project: Credit Risk Prediction

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Abstract—This paper evaluates machine learning models like Neural Networks, KNN, and Logistic Regression for predicting credit defaults on an imbalanced dataset of 45,000 records. After preprocessing and addressing class imbalance via stratified splitting, the supervised models were evaluated alongside an unsupervised K-Means Clustering. The Neural Network outperformed the others, achieving an AUC of 0.9672 and F1-Score of 0.7652, proving highly effective at finding financial risk patterns.

Keywords—Credit risk, machine learning, neural networks, data preprocessing, KNN.

I. INTRODUCTION

Financial institutions lose significant capital to defaulted loans. This project aims to build a machine learning pipeline capable of predicting credit risk. The motivation is to automate the risk assessment process, allowing banks to accurately flag high-risk applicants while safely approving low risk individuals.

The Selected Feature with discrete values as our target feature is loan_status.

II. DATASET DESCRIPTION

A. Number of Features

The raw dataset contains 13 input features and 1 target feature (14 columns total), after encoding and feature selection in preprocessing, this becomes 20 features.

B. Number of Data Points

There are 45,000 data points.

C. Types of Features

- **Categorical**
Discrete attributes such as person_gender, person_education, person_home_ownership, loan_intent and previous_loan_defaults_on_file.
- **Numerical**
Continuous variables such as person_age, person_income, person_emp_exp, loan_amnt, loan_int_rate, loan_percent_income, cb_person_cred_hist_length and credit_score.

III. DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the distribution of the dataset.

A. Numerical Feature Distributions

Histograms (Fig. 1) show person_income is heavily right skewed, along with person_age, person_emp_exp, and cb_person_cred_hist_length.

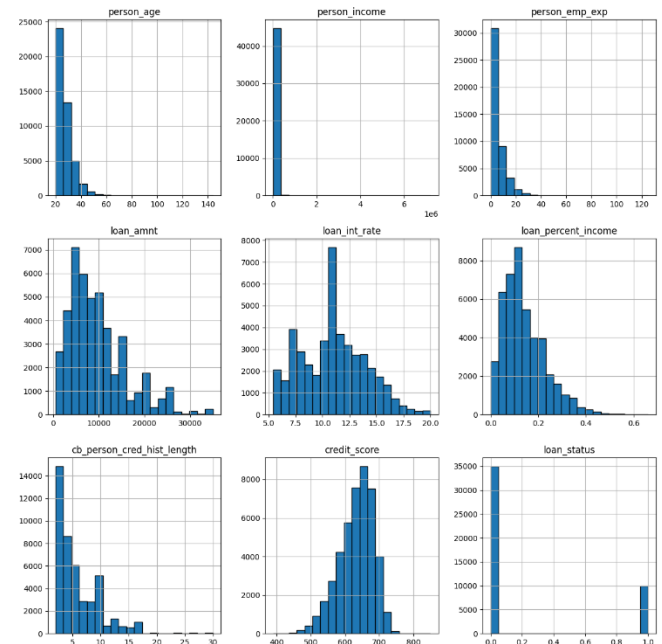


Fig. 1. Histograms of numerical features.

Boxplots (Fig. 2) revealed some impossible values: ages above 100 and employment experience above 60 years.

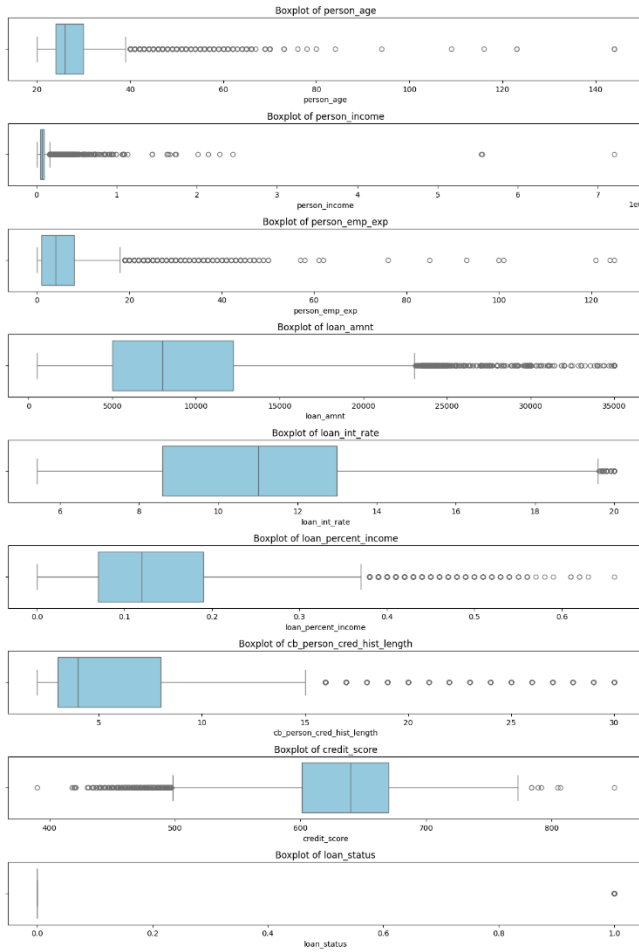


Fig. 2. Boxplots of numerical features.

B. Correlation Analysis

A correlation heatmap (Fig. 3) shows `loan_percent_income` and `loan_int_rate` have the strongest correlation with `loan_status`, while `person_age`, `person_emp_exp`, and `cb_person_cred_hist_length` are highly correlated with each other (multicollinearity).

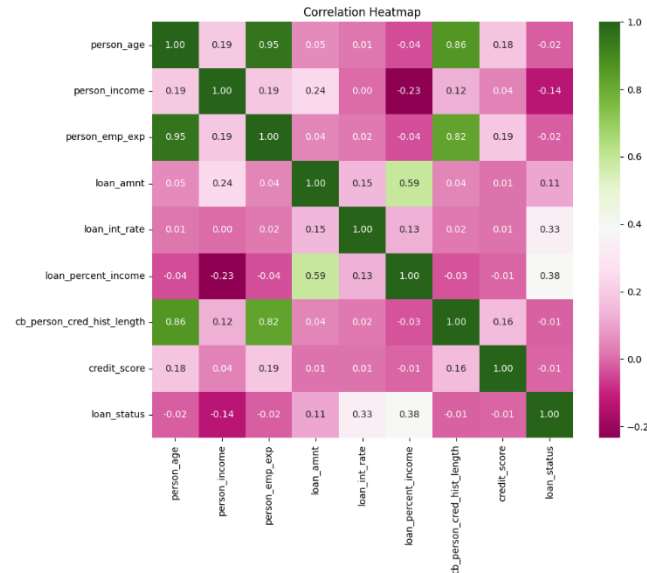


Fig. 3. Correlation heatmap of numerical features.

C. Categorical Feature Distributions

`previous_loan_defaults_on_file` (Fig. 4) was found to be the most predictive categorical feature.

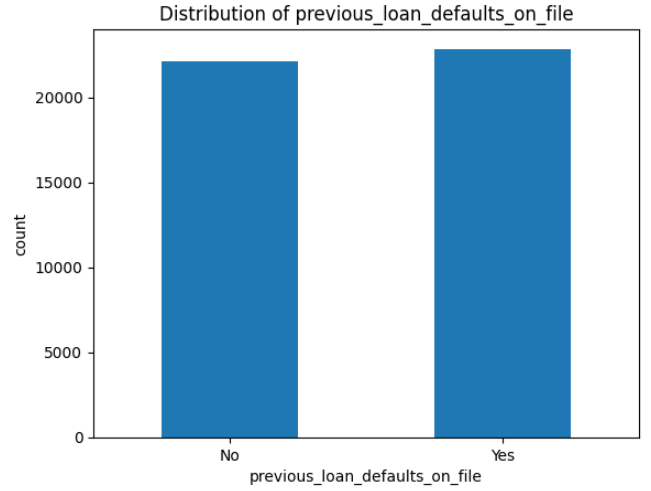


Fig. 4. Distribution of `previous_loan_defaults_on_file`.

IV. DATASET PRE-PROCESSING

A. Problem: Outliers and Invalid Values

The dataset had no missing values and no duplicate rows. However, EDA revealed impossible values: ages above 100 and employment experience exceeding age minus 16. These 10 rows were dropped, reducing the dataset to 44,990 rows.

B. Skewness and Encoding

`person_income` was log-transformed ($\log_{10}$) to correct its extreme right skew. All categorical features were one-hot encoded, which expanded the feature set to 22 columns.

C. Feature Selection

`person_emp_exp` and `cb_person_cred_hist_length` were dropped due to high correlation with `person_age` ($r = 0.95$ and 0.86), reducing the feature set to 20.

D. Feature Scaling

The data was split into training (80%) and testing (20%) sets using stratified sampling. `StandardScaler` was fit on the training set only and applied to both sets, giving 35,992 training and 8,998 testing samples.

V. MODEL TRAINING

A. Logistic Regression

A Logistic Regression classifier was trained with `class_weight='balanced'` to tackle the 78/22 class imbalance.

B. K-Nearest Neighbors (KNN)

A grid search over $k = 3$ to 21 using 5-fold cross-validation selected $k = 9$ as optimal, based on F1-score.

C. Neural Network

A Neural Network was built the following architecture:

- Dense layer, 32 units, ReLU activation

- Dense layer, 16 units, ReLU activation
- Dense (output) layer, 1-unit, Sigmoid activation

The model was trained with the Adam Optimizer, binary cross-entropy loss, class weighting for 20 epochs.

D. K-Means Clustering (Unsupervised)

K-Means was applied to the scaled features without using loan_status. The elbow method and silhouette score (Fig. 5) were computed for $k = 2$ to 10; the silhouette score peaked at $k = 5$ (0.142).

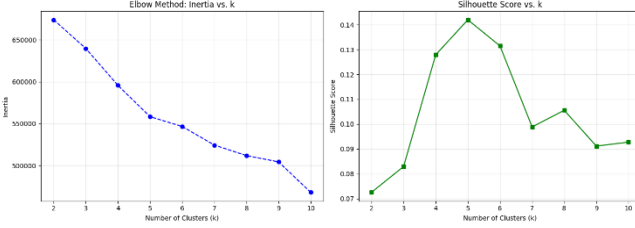


Fig. 5. Elbow method and silhouette score vs. number of clusters k .

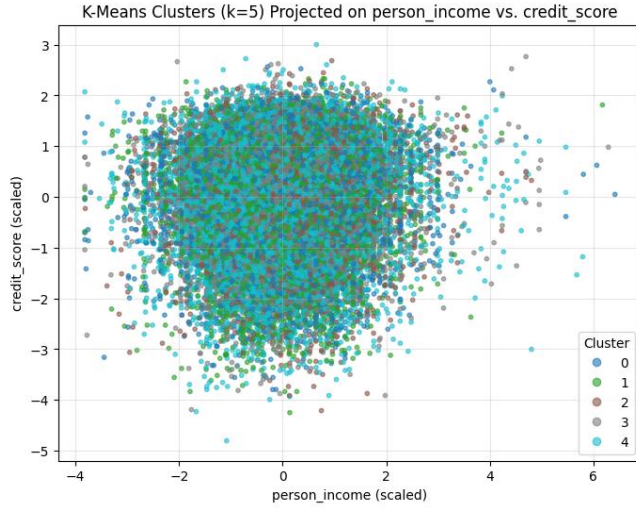


Fig. 6. K-Means Cluster ($k = 5$) projected on person_income vs credit_score

VI. MODEL EVALUATION

A. Confusion Matrices

Fig. 7 shows the confusion matrices for the three supervised models on the 8,998-sample test set.

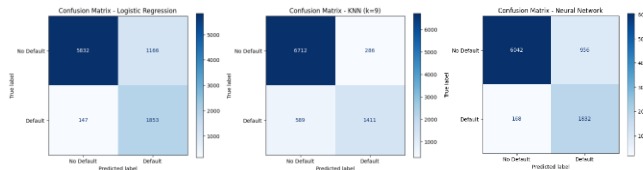


Fig. 7. Confusion matrices for Logistic Regression, KNN, and Neural Network.

B. Accuracy, Precision, Recall, F1-Score

Table I summarizes the classification metrics for each model.

TABLE I
Classification Performance on the Test Set

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8541	0.6138	0.9265	0.7384
KNN (k=9)	0.9028	0.8315	0.7055	0.7633
Neural Network	0.8751	0.6571	0.9160	0.7652

C. ROC Curve and AUC Score

Fig. 8 shows the ROC curves and AUC scores. The Neural Network has the highest AUC (0.9672), followed by Logistic Regression (0.9550) and KNN (0.9457), confirming the Neural Network best performer.

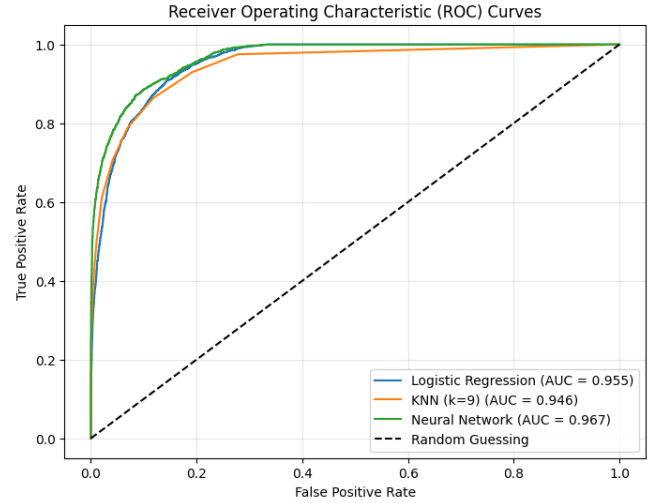


Fig. 8. ROC curves comparing Logistic Regression, KNN, and Neural Network.

VII. CONCLUSION

Based on the evaluation metrics, the Neural Network achieved the best overall performance ($F1 = 0.7652$, $AUC = 0.9672$) with strong recall (0.9160). Logistic Regression achieved the highest recall (0.9265) but lower precision, while KNN achieved the highest precision (0.8315), but lower recall compared with the other two models.

The Neural Network performed best because it can learn more complex patterns in the data than a simple straight-line model like Logistic Regression. KNN has no way to tackle the class imbalance issue, so it predicts the safer majority outcome, that's why it misses more defaulters despite scoring well on accuracy. The weak K-Means result also shows that defaulters and non-defaulters do not form clearly separate groups on their own.